Title	Authors	Abstract	EXTRA

Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms	Zander S Venter 1, Tirthankar Chakraborty 2, Xuhui Lee	**Abstract:** The ubiquitous nature of satellite data has led to an explosion of studies on the surface urban heat island (SUHI). Relatively few have simultaneously used air temperature measurements to compare SUHI with the canopy UHI (CUHI), which is more relevant to public health. Using crowdsourced citizen weather stations (>50,000) and satellite data over Europe, we estimate the CUHI and SUHI intensity in 342 urban clusters during the 2019 heat wave. Satellites produce a sixfold overestimate of UHI relative to station measurements (mean SUHI 1.45° C; CUHI 0.26°C), with SUHI exceeding CUHI in 96% of cities during daytime and in 80% at night. Using empirical evidence, we confirm the control of aerodynamic roughness on UHI intensity, but find evaporative cooling to have a stronger overall impact during this time period. Our results support urban greening as an effective UHI mitigation strategy and caution against relying on satellite data for urban heat risk assessments.	**Discussion:** While the impact of UHI on heat stress is only one of its many impacts, it has been the focus of a large number of studies, especially those looking at interactions between larger-scale phenomena like regional heat stress or global warming and urbanization (30–32). Studies on both individual (17, 33) and multiple cities (16) have shown that the SUHI is usually higher than the CUHI, especially during the daytime, which we confirm by scaling this comparison up to over 342 European cities during a heat wave period. The notably higher magnitude of SUHI has important implications for understanding and attributing the source of heat stress in urban areas. Because air temperature is the major constituent of heat stress, urban-rural air temperature differentials can help isolate the contribution of urbanization to local-scale heat exposure. Epidemiological studies have relied on near-surface air (ambient) temperature measured at fixed sites to quantify thermal exposure (34). However, spatially explicit LST data from satellites are gaining popularity and are being increasingly applied, either directly or for predicting air temperature, to public health risk studies (35, 36). In doing so, researchers run the risk of overestimating the magnitude of temperature exposure as well as the contribution of urbanization to the local temperature. This is particularly concerning given that the upward bias of satellite-based measures of UHI was greatest during conditions that are most relevant to heat risk epidemiology (daytime, low cloud cover, high diurnal temperature ranges, and low wind speeds). Proper quantification of the public health implications of UHI requires dense monitoring networks in cities, and in the absence of these monitoring networks, crowdsourced air temperature measurements can be a reasonable alternative, as we demonstrate in the present study. We also find that evapotranspiration and roughness were important determinants of both SUHI and CUHI intensity during this heat wave period (fig. S6). In particular, for the goal of reducing the effect of UHI on human health, urban green spaces remain an important mitigation option given that evapotranspiration was the dominant driver of UHI during the day (Fig. 3). However, urban greening may lead to elevated humidity levels, which is another contributor to human heat

Evaluating contributions of urbanization and global climate change to urban land surface temperature change: a case study in Lagos, Nigeria	Liying Guo, Lipingg Di, Chen Zhang, Li Lin, Fei Chen, Alamin Molla	**Abstract:** This study aims to develop a general method and evaluate the contributions of localized urbanization and global climatechange to long-term urban land surface temperature (ULST) change. Combined daytime and nighttime daily MODISproducts were used to ll data biases from satellite-observed data and applied to tropical regions during dry season from2003 to 2021. Lagos, a fast-growing metropolis in West Africa, is selected as an example for reference. The results aresummarized as follows. 1) Long-term annual ULST is conrmed to be controlled by urbanization, global climate change,and interannual climate variation. 2) Linear regression is used to remove the interannual climate uctuations on long-termobserved LST time series. 3) Urban-rural comparison method has been developed to separate the contributions ofurbanization and global climate change to long-term ULST changes. Both urbanization and global climate change areveried to contribute to the ULST increase with positive trends. 4) Daytime ULST increased the most in the afternoon time ata mean rate of 1.429 ºC per decade, with 0.985 ºC (10yr)-1 contributed by urbanization and 0.444 ºC (10yr)-1 contributed byglobal climate change. Nighttime ULST increased the most after midnight at a rate of 0.563ºC (10yr) -1, with 0.56 ºC (10yr)-1 contributed by urbanization and 0.003 ºC (10yr) -1 contributed by global climate change. 5) Urbanization is generally tobe responsible for around 60.97% of the urban warming. The increasing urban-induced hot has been causing urbanresidents to suffer more heat-based health risks and increasing the extreme events. Therefore, the results are of usefulreference for both urbanization and climate change related issues in the geo-science field	

Thermal changes along the urban-rural continuums in Southeast Asia	Yuanhui Zhu*, Soe W Myint, Jiquan Chen, Peilei Fan, Karen C Seto, Atul Jain, Jiaguo Qi and Jin Wang	**Abstract:** Southeast Asia's (SEA) rapid urbanization has intensified urban heat effects. While previous studies address urban warming at specific time points, the dynamic relationship between land cover and land use (LCLU) changes and land surface temperature (LST) along urban-rural continuums remains understudied. This study examines LST, LCLU, and NDVI changes across 19 SEA cities of varying urbanization levels from 2000 to 2022 using Landsat images. Results show that LST change rates differ by urbanization levels, with built-up lands driving urban warming. Larger cities exhibited more intense and widespread LST increases extending into rural areas. The megacities experienced significant LST growth within 5 km of city centers despite limited built-up expansion. However, some areas in many cities experience decreasing trends of heat. NDVI significantly influenced LST changes through its cooling effect and the warming impact due to vegetation loss caused by urban expansion. Findings support urban planning strategies that balance growth with enhancing green spaces, redirecting growth to smaller cities, and shifting development from city centers to urban outskirts.	

Integrating Satellite-Based and Participatory Spatial Analyses for Local, Contextualised, and Nuanced Extreme Heat Hotspot Mapping in Overheated Urban Southeast Asia	aurence L. Delina, Homer Pagkalinawan, Sharon Feliza Ann Macagba	**Abstract:** Rapid urbanisation in Southeast Asia intensifies extreme-heat exposure, especially where high population density, extensive impervious surfaces, and limited green cover coincide. This study combines multi-decadal satellite-derived land surface temperature (LST) mapping with spatial regression and community participatory mapping to locate community-level heat clusters of extreme heat in Metro Manila and the Bangkok Metropolitan Area. Using a 1 km grid and a suite of built-environment, infrastructural, natural feature, and socioeconomic covariates (including two policy-oriented efficiency ratios), we find that: (1) LST clusters exhibit significant positive spatial autocorrelation, and (2) a statistically significant quadratic relationship links a building-efficiency ratio to local temperatures in both cities—implying an optimal level of compactness beyond which further densification reduces surface heat (consistent with increased shading and compact-form benefits). In addition, higher population density, dense road networks, and proximity to extensive heat-retaining infrastructure (notably airports) are consistently associated with elevated temperatures, whereas greater greenspace accessibility is associated with cooling. Crucially, participatory mapping supplied fine-scale, lived-experience data that (a) identified microhotspots that LST products missed, (b) revealed exposure pathways linked to travel distances and service access, and (c) highlighted social vulnerabilities and coping practises that inform equitable intervention priorities. Together, the integrated approach yields place-sensitive insights—translating statistical relationships into implementable targets (e.g., green space per capita, building-to-economic activity benchmarks) for targeted greening, shading, and accessibility improvements. We discuss limitations—namely, the 1 km LST resolution and the small participatory sample—and argue that scaling this framework will better guide locally appropriate, equity-centred heat adaptation in megacities.	

Satellite Remote Sensing of Surface Urban Heat Islands: Progress, Challenges, and Perspective	Decheng Zhou , Jingfeng Xiao , Stefania Bonafoni , Christian Berger, Kaveh Deilami,Yuyu Zhou ,Steve Frolking ,Rui Yao,Zhi Qiao and José A. Sobrino	**Abstract:** The surface urban heat island (SUHI), which represents the difference of land surface temperature (LST) in urban relativity to neighboring non-urban surfaces, is usually measured using satellite LST data. Over the last few decades, advancements of remote sensing along with spatial science have considerably increased the number and quality of SUHI studies that form the major body of the urban heat island (UHI) literature. This paper provides a systematic review of satellite-based SUHI studies, from their origin in 1972 to the present. We find an exponentially increasing trend of SUHI research since 2005, with clear preferences for geographic areas, time of day, seasons, research foci, and platforms/sensors. The most frequently studied region and time period of research are China and summer daytime, respectively. Nearly two-thirds of the studies focus on the SUHI/LST variability at a local scale. The Landsat Thematic Mapper (TM)/Enhanced Thematic Mapper (ETM+) /Thermal Infrared Sensor (TIRS) and Terra/Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) are the two most commonly-used satellite sensors and account for about 78% of the total publications. We systematically reviewed the main satellite/sensors, methods, key findings, and challenges of the SUHI research. Previous studies confirm that the large spatial (local to global scales) and temporal (diurnal, seasonal, and inter-annual) variations of SUHI are contributed by a variety of factors such as impervious surface area, vegetation cover, landscape structure, albedo, and climate. However, applications of SUHI research are largely impeded by a series of data and methodological limitations. Lastly, we propose key potential directions and opportunities for future efforts. Besides improving the quality and quantity of LST data, more attention should be focused on understudied regions/cities, methods to examine SUHI intensity, inter-annual variability and long-term trends of SUHI, scaling issues of SUHI, the relationship between surface and subsurface UHIs, and the integration of remote sensing with field observations and numeric modeling.	**Summary:** This review shows an exponentially increasing trend in satellite-based SUHI publications since 2005, with large biases in the geographic areas, study time of day, study seasons, and research foci. Local-scale SUHI variations in the daytime and/or summer have received more research attention. Knowledge of SUHI variations and their relationship with air UHIs has been greatly enriched by the past SUHI studies even though some findings such as the relative significance of landscape composition, configuration, and the dominant factors for SUHI variations among cities remained highly disputed. UHI researchers currently face a series of challenges in using satellite LST data, including the differences between satellite-derived LST and air temperature, the impacts of clouds and other factors on LST data, the trade-off between spatial and temporal resolutions, the methods to quantify SUHI intensity, concurrent land cover, land use mapping, accuracy assessment, and attribution of SUHI. Worse still, the resultant uncertainties have been magnified by the wide applications of very limited datasets (e.g., using one image in a season or year) in the SUHI analyses. Given the availability of long-term archive/free data, a large number of images or time-series analyses are strongly recommended for future studies to increase the robustness of the findings. Considerable scope for improving the quality and quantity of LST data and analysis remains. We suggest that more attention be paid to the understudied regions/cities, the methods to calculate SUHI intensity, inter-annual variability and long-term trends of SUHI, scale issues of SUHI, SUHI-SubUHI relationships, and the integration of remote sensing data with field observation and numeric modeling in the future.

Satellite-derived land surface temperature: Current status and perspectives	Zhao-Liang Li , Bo-Hui Tang , Hua Wu , Huazhong Ren , Guangjian Yan , Zhengming Wan , Isabel F. Trigo , José A. Sobrino	**Abstract:** Land surface temperature (LST) is one of the key parameters in the physics of land surface processes from local through global scales. The importance of LST is being increasingly recognized and there is a strong interest in developing methodologies to measure LST from space. However, retrieving LST is still a challenging task since the LST retrieval problem is ill-posed. This paper reviews the current status of selected remote sensing algorithms for estimating LST from thermal infrared (TIR) data. A brief theoretical background of the subject is presented along with a survey of the algorithms employed for obtaining LST from space-based TIR measurements. The discussion focuses on TIR data acquired from polar-orbiting satellites because of their widespread use, global applicability and higher spatial resolution compared to geostationary satellites. The theoretical framework and methodologies used to derive the LST from the data are reviewed followed by the methodologies for validating satellite-derived LST. Directions for future research to improve the accuracy of satellite-derived LST are then suggested.	
Surface warming in global cities is substantially more rapid than in rural background areas	Zihan Liu, Wenfeng Zhan, Benjamin Bechtel, James Voogt, Jiameng Lai, Tirthankar Chakraborty, Zhi-Hua Wang, Manchun Li, Fan Huang & Xuhui Lee	**Abstract:** Warming trends in cities are influenced both by large-scale climate processes and by local-scale urbanization. However, little is known about how surface warming trends of global cities differ from those characterized by weather observations in the rural background. Here, through statistical analyses of satellite land surface temperatures (2002 to 2021), we find that the mean surface warming trend is 0.50 ± 0.20 K·decade−1 (mean ± one S.D.) in the urban core of 2000-plus city clusters worldwide, and is 29% greater than the trend for the rural background. On average, background climate change is the largest contributor explaining 0.30 ± 0.11 K·decade−1 of the urban surface warming. In city clusters in China and India, however, more than 0.23 K·decade−1 of the mean trend is attributed to urban expansion. We also find evidence of urban greening in European cities, which offsets 0.13 ± 0.034 K·decade−1 of background surface warming.	

Reexamining the relationship between surface urban heat island intensity and annual precipitation: Effects of reference rural land cover	Xiaoshan Yang, Lingye Yao	**Abstract:** Previous studies have shown that surface urban heat island intensity (SUHII) across cities is significantly positively correlated with mean annual precipitation (MAP), including linear and nonlinear correlations. Different explanations for these SUHII–MAP relations have been suggested, yet a systematic examination of the impact of variability in reference rural land cover on geographic variations of daytime SUHII and the relation with MAP is still lacking. In this study, the previously proposed SUHII–MAP relations are reexamined by investigating 60 cities across North America and 346 cities across the globe, respectively. The focus of the study is on the role of variability of reference land cover types in the SUHII–MAP relations. The 10-yr time series of satellite-observed land surface temperature, precipitation and solar radiation data, in conjunction with global land cover data, are used for analysis. Our results confirm the reproducibility of both the linear and nonlinear SUHII–MAP relations; nevertheless, the significant positive correlations between the daytime SUHII and the MAP are found to be dependent mainly on the variability of the reference land cover types associated with MAP levels (i.e., varying from desert to tree cover). In contrast, when a constant land cover type is taken as the reference, no significant correlations between the daytime SUHII and the MAP are observed. The finding highlights that respecting SUHI intensity as a relative measure and accounting for variability of reference land cover is of crucial importance for proper interpretation and understanding of SUHI.	